

APARNA ANANTHASUBRAMANIAM

Ph.D. Student

Information and Social Work, University of Michigan

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EDUCATION

University of Michigan

- Ph.D., Information (advisor: Daniel M. Romero) 2019 – present
- Ph.D., Social Work and Social Welfare (advisor: Kristin S. Seefeldt) 2024 – present
- M.S.W., Social Work 2024 – present
concentration: Interpersonal Practice in Health, Mental Health, and Substance Use

Stanford University

- M.S., Management Science & Engineering 2013 – 2014
concentration: Operations and Analytics
- B.S., Mathematics 2009 – 2014

PEER-REVIEWED PUBLICATIONS

* denotes equal contributions, † denotes lead analyst (non-first-author, led execution and writing of the paper), ° denotes a student I mentored

Socioeconomic Shocks + Macro and Meso Contexts:

- [1] **Ananthasubramaniam A**, Thulin E, Kalesnikava V, Falde S°, Kertawidjaja J°, Johns L, Rodriguez-Putnam A, Spring E, Zivin K, Mezuk B. (2026) “Characterizing Online Activities Contributing to Suicide Mortality Among Youth.” accepted at *International AAAI Conference on Web and Social Media (ICWSM)*.
Third place (top ~8%, out of 42 submissions) in the CDC Youth Mental Health: Novel Variables Data Challenge
Presented at: International Conference on Computational Social Science (oral, 2025)
- [2] Mezuk B, Kalesnikava VA*†, **Ananthasubramaniam A***†, Lane A, Rodriguez-Putnam A, Johns L, Bagge C, Burgard S, Zivin K. (2024) “Suicide in the US during the First Year of the Pandemic: Evidence from the National Violent Death Reporting System.” *PLoS One*. 19 (10), e0312027.
Presented at: Michigan Language and Speech Days (2024, poster); International Conference on Computational Social Science (2024, poster)
- [3] **Ananthasubramaniam A**, Jurgens D, Kahsay E, Mezuk B. (2024) “Retirement-Related Suicide during the Great Recession: Perspectives from the National Violent Death Reporting System.” *The Gerontologist* 64(6).
Presented at: Association for Public Policy Analysis and Management (2025, poster), National Violent Death Reporting System Conference (2023, oral); Michigan Language

and Speech Days (2024, poster); International Conference on Computational Social Science (2024, poster)

- [4] Dang L*, **Ananthasubramaniam A***, Mezuk B. (2022) “Spotlight on the Prevalence and Challenges of Depression following Retirement: Opportunities for interventions.” *Clinical Interventions in Aging*, 1037-1056.

Social Networks and Social Influence:

- [5] (random order of first authors) Xu N*, Chen H*, Rakshit S*, **Ananthasubramaniam A***, Yada O*, Zheng M*, Jiang M*, Zhang L*, Yi B*, Alkeik K*, Israeli A*, Shu B*, Shen H*, Pei J*, Zhang H*, Jurgens D. (2025) “Causally Modeling the Linguistic and Network Factors that Predict Email Response.” *Proceedings of the 2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*.
Presented at: International Conference on Computational Social Science (2025, poster)
- [6] **Ananthasubramaniam A**, Zhu Y°, Jurgens D, Romero DM. (2025) “The Role of Network and Identity in the Diffusion of Hashtags.” *Proceedings of the Web Conference (WWW)*.
Presented at: International Conference on Computational Social Science (2024, oral); Contagion on Complex Social Systems Workshop (2023, oral)
- [7] **Ananthasubramaniam A**, Jurgens D, Romero DM. (2024) “Networks and Identity Drive Geographical Properties of the Diffusion of Linguistic Innovation.” *npj (Nature Partner Journal) Complexity*, 1(14).
Presented at: Contagion on Complex Social Systems Workshop (2023, oral); International Conference on Computational Social Science (2022, poster); Data Science / Computational Social Science Seminar, University of Michigan School of Information (2022, oral); International Conference on Computational Social Science (2020, oral)

Misc:

- [8] Pei J, **Ananthasubramaniam A**, Wang X, Zhou N, Sargent J, Jurgens D. (2022) “Potato: The Portable Annotation Tool.” *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*.
Presented at: AAAI Conference on Human Computation and Crowdsourcing (2024, demo by JP) – received Best Demo Paper Award
- [9] Diniz Behn C, **Ananthasubramaniam A**, Booth V. (2013) “Contrasting existence and robustness of REM/NREM cycling in physiologically-based models of regulatory networks.” *SIAM Journal of Applied Dynamical Systems*, 12(1), 279–314
Presented at: Northern California Undergraduate Mathematics Conference (2011, oral)

WORKSHOPS PAPERS (ARCHIVAL, LIGHTLY PEER-REVIEWED):

- [10] (random order of first authors) **Ananthasubramaniam A***, Chen H*, Yan J*, Alkeik K*, Pei J*, Seth A*, Dunagan L*, Choi M*, Litterer B*, Jurgens D. (2023) “Exploring

Linguistic Style Matching in Online Communities: The Role of Social Context and Conversation Dynamics.” *Proceedings of the First Workshop on Social Influence in Conversations (SICon 2023)*, 64–74, Toronto, Canada. Association for Computational Linguistics.

Best Paper Award (top submission)

MANUSCRIPTS PENDING PEER REVIEW:

- [11] **Ananthasubramaniam A**, Israeli A, Jurgens D, Romero DM. (2025) “Layers of Influence: Modeling the Effects of Relationships in Information Diffusion using Multi-Layer Networks.” under review (manuscript available upon request).
- [12] (random order of first authors) Ivey J*, Kumar S*, Liu J*, Shen H*, Rakshit S*, Raju R*, Zhang H*, **Ananthasubramaniam A***, Kim J*, Yi B*, Wright D*, Israeli A*, Møller AG*, Zhang L*, Jurgens D. (2025) “Real or Robotic? Assessing Whether LLMs Accurately Simulate Qualities of Human Responses in Dialogue.” under review ([arXiv](#)).
Presented at: International Conference on Computational Social Science (2025, poster)
- [13] Zhong C^{o†}, Danh L, **Ananthasubramaniam A[†]**, Mezuk B. (2024) “Understanding the Links between Social Isolation and Functional Limitations: A Network Analysis.” under review.
Presented at: Gerontological Society of America Conference (2024, oral by CZ)
- [14] **Ananthasubramaniam A**, Burgard S, Romero DM, Jurgens D, Mezuk B. (2025) “State Policies Moderate Risk of Suicide Death Following Eviction and Foreclosure.” in prep (manuscript available upon request).
Presented at: Data Science Rising Stars (2025, oral), Association for Public Policy Analysis and Management (2025, oral), International Conference on Computational Social Science (2024, poster); Michigan Institute for Clinical and Health Research IDEAL Symposium (2024, poster); Michigan Language and Speech Days (2024, poster)
- [15] **Ananthasubramaniam A**, Quinonez J, Seefeldt KS. (2025) “Material Hardship among Immigrants Left out of the CARES Act: a Comparative Analysis of California and Texas in 2020.” in prep (manuscript available upon request).
Presented at: Association for Public Policy Analysis and Management (2025, poster)

AWARDS

- [1] Rising Star in Data Science, 2025 (*awarded to 31 senior PhD students and postdocs in the US, by data science faculty at Stanford University, University of Chicago, and University of California San Diego*)
- [2] Honorable Mention, UMSI Gary M. Olson Outstanding Ph.D. Student of the Year, 2025 (*one award and two honorable mentions out of all the department’s 50+ PhD candidates*)

- [3] Third place, CDC Youth Mental Health: Novel Variables Data Challenge, 2024 (*out of 42 entries, top 8%*)
- [4] UMSI Outstanding Graduate Student Instructor of the Year Award, 2023 and 2024 (*given to one PhD student each year; in 25 years, I am only the second student to have won this award multiple times; also nominated in 2022*)
- [5] Best Paper Award, Workshop on Social Influence in Conversations, 2023 (*given to the top submission, based on reviewer and organizer assessments*)
- [6] UMSI Team Award for Impact in Diversity, Equity, & Inclusion, 2022 (*for organizing the Race and Technology Reading Group; given to one PhD student initiative each year*)
- [7] US Presidential Scholar, US Department of Education, 2009 (*selected as one of two scholars among high school seniors from the state of Michigan*)

TECHNICAL REPORTS (NOT PEER-REVIEWED):

- [1] Mission Asset Fund. (2021). MAF's Financial Equity Framework: A Solution Rooted in Community.
- [2] Mission Asset Fund. (2021). COVID-19's Impact on Immigrant Families.
- [3] Mission Asset Fund. (2021). COVID-19's Impact on California Public College Students.
- [4] Mission Asset Fund. (2019). Invisible Barriers: Navigating Financial Services with an ITIN
- [5] Mission Asset Fund. (2019). New Insights from Working with Entrepreneurs.
- [6] Mission Asset Fund. (2018). Insights from a Survey of 5,000 DACA Recipients.

CONFERENCE PRESENTATIONS (LIGHTLY PEER-REVIEWED, NON-ARCHIVAL):

- [1] Modeling Tradeoffs of Security Risks in Telemetric Cardiac Pacemakers. International Conference on Computational Social Science (2020, poster)
- [2] Agent-Based Modeling Workshop. Active Living Research Conference (2014, oral)

TEACHING

University of Michigan Teaching (as a PhD student):

- [1] SI 301: Models of Social Information Processing (Lead Graduate Student Instructor), Fall 2024
- [2] SI 301: Models of Social Information Processing (Lead Graduate Student Instructor), Fall 2023, **UMSI Outstanding Graduate Student Instructor of the Year** (award given to 1 PhD student and 1 MSI student)
- [3] SI 301: Models of Social Information Processing (Graduate Student Instructor), Fall 2022, **UMSI Outstanding Graduate Student Instructor of the Year** (award given to 1 PhD student and 1 MSI student)
- [4] SI 544: Introduction to Statistics and Data Analysis (Graduate Student Instructor), Fall 2021

Stanford University Teaching (as an undergraduate student):

- [5] Phil 197I: Homelessness in Silicon Valley (Student-Led Course, Course Creator and Mentor to Student Instructors), Winter 2015 - 2017
- [6] Math 19-21: Single Variable Calculus (Undergraduate Teaching Assistant), AY 2011, 2013 – 2014
- [7] Peds 299: Implementing Behavior Change Interventions (Student-Led Course, Creator and Instructor), Winter 2013
- [8] EarthSys 18: Designing Behavior Change Interventions (Student-Led Course, Instructor), Fall 2011 – 2013

University of Michigan Guest Lectures (as a PhD student):

- [1] “Bayes’ Theorem” for SI 301: Models of Social Information Processing, Fall 2023
- [2] “Logistic Regression and Regression Discontinuity,” SI 330: Data Manipulation, Winter 2023
- [3] “Spatial Data Analysis,” SI 330: Data Manipulation, Winter 2023
- [4] “Information Retrieval Applications” for SI 650: Information Retrieval, Fall 2022
- [5] “Structural Balance Theorem,” SI 301: Models of Social Information Processing, Fall 2022
- [6] “Causal Inference Overview,” SI 544: Introduction to Statistics and Data Analysis, Fall 2021

STUDENTS

- [1] Genna Ashymov, 2025-present (MSW, Social Work, University of Michigan)
- [2] Lia Du, 2025-present (BSI, Information, University of Michigan)
- [3] Silas Falde, 2024-2025 (BS, Data Science in Engineering, University of Michigan)
- [4] Jonathan Kertawidjaja, 2024-2025 (BS, Data Science, University of Michigan)
- [5] Chuwen (Cullen) Zhong, 2023-2025 (MPH, Public Health, University of Michigan → PhD at University of Michigan School of Public Health)
- [6] Ziheng Xu, 2023-2024 (BS, Economics and Cognitive Science, University of Michigan → MS at Cornell University)
- [7] Yufei (Louise) Zhu, 2022-2023 (MS, Data Science in Engineering, University of Michigan → PhD at George Washington University)

ACADEMIC SERVICE (at University of Michigan)

Student Representative, Curriculum Committee, School of Social Work Fall 2025 – present
Served as the student representative on the School of Social Work’s curriculum committee, which was charged with evaluating pathways and course offerings for the BSW and MSW programs.

Assistant, Social Work Grand Rounds Fall 2025 – present
Supported the School of Social Work’s Associate Dean for Research in coordinating Social Work Grand Rounds on the topic of AI in Social Work.

Founder / Co-Organizer, Computational Social Science Working Group Win 2023 – Sum 2024
Created a cross-disciplinary space for researchers studying social questions using computational methods. Built a mailing list of over 200 graduate students, research staff, and faculty from nearly 20 departments. Held biweekly working group meetings for research presentations (~15-20 attendees from 2-4 departments each meeting) and semesterly mixers for networking (~55-65 attendees from 10+ departments each event). Managed annual budget of \$1500.

Co-Organizer, Race and Technology Reading Group Fall 2020 – Sum 2023
Held weekly meetings for faculty, staff, and students at the School of Information to discuss books about race and racism in technology (~5-10 attendees per week). Developed antiracist facilitation materials that allowed participants to share lived experience in academic discussions. Invited 3 faculty members to discuss their books with the group. Managed \$3000 annual budget and mailing list of over 250 people.

UMSI Team Award for Impact in Diversity, Equity, & Inclusion (2022, given to 1 PhD-student-led initiative)

Member, Committee on Doctoral Student Workspace Allocations Win 2020, 2021
Worked with the School of Information’s Academic Dean and Doctoral Executive Committee to determine how to assign limited departmental workspaces to PhD students. Authored and analyzed a survey (81% response rate) to develop principles for allocation in a member-engaged way. Supported the implementation of the new allocation system, which remained in use by department administration until the school moved to a larger building in 2025.

NON-ACADEMIC WORK EXPERIENCE AND SERVICE

Nonprofit Research and Technology:

Researcher (Contract), Mission Asset Fund, San Francisco, CA 2020 – present
Currently consulting on design of and lead analyst for a 2-year study of guaranteed income for immigrant families recovering from the COVID-19 recession. Developed a “Financial Equity Framework” to prioritize the waitlist of a \$25M Rapid Response Fund, giving cash assistance grants to college students, Los Angeles Creative Economy workers, and immigrants left out of CARES Act funding. Launched three bilingual surveys to measure financial impacts of COVID-19 for grant recipients’ and their financial coping strategies. Developed data-driven insights to

policymakers, including in CA's State Legislature. Developed models to measure the financial wellbeing of low-income immigrants.

Research Director, Mission Asset Fund, San Francisco, CA 2018 – 2019

Conducted user research via interviews, focus groups, surveys, and analysis of client credit reports to inform the design of technology and help tell a nuanced, strengths-based story about the financial wellbeing of low-income and immigrant communities in USA. Developed research insights presentations on entrepreneurship, informal income, credit tradelines, ITIN services, and DACAmented youth for funders and policymakers.

Research & Development Director 2016 – 2018

Research & Development Manager 2015 – 2016

Managed a fintech nonprofit's research and technology work, including supervising a team of up to 3 employees and 4 remote engineers. Developed operations plans, data systems, and front-end interfaces for many business-critical processes, including loan management, accounting, underwriting, application review, and funder/partnership management. Developed the first version of a mobile app to help our clients track progress in our programs, view financial education content, and set and track progress towards goals. When DACA was rescinded in 2017, led the development of business processes and technology to increase our throughput by 20x, so we could offer fee assistance for renewal costs to over 5K recipients in under 1 month.

Product & Research Fellow 2014 – 2015

Authored a survey to measure impact and help stakeholders understand our clients' financial lives. Developed data workflows and dashboards to report on the organization's key outcomes.

Data Analyst, Community Technology Alliance, San Jose, CA 2012 – 13

Supported five Bay Area counties in understanding their unhoused residents' demographics, housing status, service use, and program successes. Collaborated on two county-wide initiatives on reducing chronic homelessness and implementing a rapid re-housing program, including understanding and forecasting retention in housing and monitoring and evaluating early results.

Interpersonal Practice (mostly unpaid, part-time staff roles):

ACT MSW Intern, Washtenaw County Community Mental Health, Ypsilanti, MI 2025 – present

Conducted MSW practicum on the county's Assertive Community Treatment (ACT) team.

Duties included providing medications, daily check-ins, emergency support, counseling, and intensive case management to adults with severe and persistent mental illness in their homes.

Served as a provider on a suicide-prevention intervention research study, using motivational interviewing to facilitate access to the 988 crisis line.

Founding Program Director, Overnight Warming Location, Palo Alto, CA 2023 – present

Partnered with the City of Palo Alto, Palo Alto Library, and Heart and Home Collaborative (see below) to open the first and only warming center in the northern part of Silicon Valley, in response to an increase in hypothermia deaths in the area. Developed all operations manuals, program policies, and staff training materials. Recruited, trained, and supervised shelter staff, including being on call for the entire duration of service. Managed community outreach to referral partners and a direct SMS alert system for clients. Created and oversaw all ancillary

services (e.g., laundry, meals, client resources, outreach, referrals). Collaborated with City government partners on supplies management, security and janitorial services, and community outreach. Presented the program to key community stakeholders including City Council and the city's Human Resources Commission and open houses for the general public.

Board of Directors, Heart and Home Collaborative, Palo Alto, CA 2016 – 19, 2022 – present
Led major fundraising initiatives, including securing a grant from the City of Palo Alto for 30-50% of our annual operating budget in 2023 and writing our first three annual reports from 2023-2025. Supervised 2 community services work study undergraduate students. Helped recruit four faith organizations to host the women's shelter. Chaired the organization's policy committee, including developing policies and operations manuals for a cash assistance program and revising and adding new programmatic and organizational policies each year. Represented the organization in city- and county-wide service provider collaboratives.

Founding Shelter Director and Client Services Director 2014 – 2019

Trainee / Shelter Worker, LifeMoves Shelter Partnership 2011 – 2013

Opened a volunteer-run shelter for 15 unhoused women. Served as one of two Shelter Directors, developing and regularly revising the shelter's operations manuals and policies using participatory planning methodologies, training and supervising staff, managing all client touchpoints (intake, discharge, conflicts, house meetings), and providing all wraparound services (housing and job search, medical support, transportation, ad hoc needs, sympathetic ear). Managed key operations including community outreach, supplies, and donations.

Case Manager/Crisis Manager, Neighbors Helping Neighbors, Palo Alto, CA 2014 – 16

Assessed needs, created action plans, helped clients navigate resources and medical/financial systems, and offered other support to low- and middle-income adults in Silicon Valley. Worked with clients with higher levels of need and who required ongoing support, including those with serious, often co-occurring housing, financial, medical, and mental health challenges.

Co-Founder and Member, Community Cooperation Team, Palo Alto, CA 2011 – 13

Started a community-led group that brought together housed and unhoused residents around local policy. Successfully petitioned Palo Alto's City Council to overturn a proposed vehicle habitation ban in 2011. Organized against another vehicle habitation ban and the closure of overnight public bathrooms in 2013. The group remained active through 2015.